Supplementary Material for MDL Meets Latent Confounders: LNML-based Causal Discovery

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A Proof of Theorem 1

Proof. Denote the values of nodes A and B as a and b, respectively. Assume

$$\begin{aligned}
&Z \sim \mathcal{N}(\mu, \sigma_0^2),\\
&a = \alpha_1 + \alpha_2 z + \epsilon_A, \quad \epsilon_A \sim \mathcal{N}(0, \sigma_1^2),\\
&b = \beta_1 + \beta_2 z + \epsilon_B, \quad \epsilon_B \sim \mathcal{N}(0, \sigma_2^2).
\end{aligned}$$

The latent relations in a and b are unobserved.

Consider the RHS of (8). The code-length $L(\cdot)$ of a node is computed by (4). Since node A has no parents, the first term becomes

$$L(A'_{\text{res}} | \text{pa}_{A'_{\text{res}}} = \emptyset) = -\log P(A),$$

where $-\log P(A)$ is the negative log-likelihood of A under the MLE

$$\begin{aligned}
\hat{\mu}_A &= \alpha_1 + \alpha_2 \mu,\\
\hat{\sigma}_A^2 &= \alpha_2^2 \sigma_0^2 + \sigma_1^2.
\end{aligned}$$

The second term $L(B'_{\text{res}} | \text{pa}_{B'_{\text{res}}} = A'_{\text{res}})$ depends on the relation between B and A. The latent linear relation can be rewritten as

$$b = \left(\beta_1 - \tfrac{\alpha_1}{\alpha_2}\beta_2\right) + \tfrac{\beta_2}{\alpha_2} a + \left(\epsilon_B - \tfrac{\beta_2}{\alpha_2}\epsilon_A\right).$$

For notational simplicity, let

$$\begin{aligned}
&b = \gamma_1 + \gamma_2 a + \epsilon'_B, \quad \epsilon'_B \sim \mathcal{N}(0, \sigma_B'^2),\\
&a \sim \mathcal{N}(\mu_A, \sigma_A^2), \quad b \sim \mathcal{N}(\gamma_1 + \gamma_2 \mu_A, \gamma_2^2 \sigma_A^2 + \sigma_B'^2).
\end{aligned}$$

Since the true mechanism between A and B is unknown, we assume $B = f(A) + \epsilon_1$ follows a Gaussian Process prior:

$$f \sim \mathcal{GP}(m(a), \kappa(a, a')).$$

Then $\boldsymbol{b}|\boldsymbol{a}$ follows a multivariate Gaussian distribution with covariance $\boldsymbol{\Sigma}_1 = \boldsymbol{K}_1 + \sigma_1^2 \boldsymbol{I}$:

$$\boldsymbol{b}|\boldsymbol{a} \sim \mathcal{N}(\boldsymbol{m}_1, \boldsymbol{\Sigma}_1).$$

The corresponding likelihood is

$$\begin{aligned}-\log P_{\text{GP}}(B|A) =& \tfrac{1}{2}(\boldsymbol{b}-\boldsymbol{m}_1)^\top \boldsymbol{\Sigma}_1^{-1}(\boldsymbol{b}-\boldsymbol{m}_1)\\ &+ \tfrac{1}{2}\log\det \boldsymbol{\Sigma}_1 + \tfrac{n}{2}\log 2\pi,\end{aligned}$$

where n is the sample size.

In theory, under a perfect GP regression and assuming the kernel has infinite capacity, the mechanism $f(\cdot)$ can recover the true relationship. In this case,

$$\begin{aligned}-\log P_{\text{GP}}(B|A) =& \tfrac{n}{2}\log\big(2\pi(\gamma_2^2\sigma_A^2 + \sigma_B'^2)\big)\\ &+ \sum_{i=1}^{n} \frac{(b_i - \gamma_1 - \gamma_2\mu_A)^2}{2(\gamma_2^2\sigma_A^2 + \sigma_B'^2)}.\end{aligned}$$

Moreover, as $\boldsymbol{K}_1 \to 0$, we have

$$L(B'_{\text{res}}|\text{pa}_{B'_{\text{res}}} = A'_{\text{res}}) \to -\log P_{\text{GP}}(B|A).$$

Thus, the RHS of (8) tends to

$$\begin{aligned}L(A'_{\text{res}}|\text{pa}_{A'_{\text{res}}} = \emptyset)+&L(B'_{\text{res}}|\text{pa}_{B'_{\text{res}}} = A'_{\text{res}})\\ &\to -\log P(A) - \log P_{\text{GP}}(B|A).\end{aligned}$$

Similarly, the LHS tends to

$$\begin{aligned}L(A'_{\text{res}}|\text{pa}_{A'_{\text{res}}} = B'_{\text{res}})+&L(B'_{\text{res}}|\text{pa}_{B'_{\text{res}}} = \emptyset)\\ &\to -\log P(B) - \log P_{\text{GP}}(A|B).\end{aligned}$$

Therefore,

$$\Delta \approx -\log P(A) - \log P_{\text{GP}}(B|A) + \log P(B) + \log P_{\text{GP}}(A|B).$$

More explicitly, the first part is

$$\begin{aligned}-\log P(A) + \log P(B) =& \tfrac{n}{2}\log\left(\tfrac{\alpha_2^2\sigma_0^2+\sigma_1^2}{\beta_2^2\sigma_0^2+\sigma_2^2}\right)\\ &+ \sum_{i=1}^{n}\left(\frac{(a_i-\alpha_1-\alpha_2\mu)^2}{2(\alpha_2^2\sigma_0^2+\sigma_1^2)} - \frac{(b_i-\beta_1-\beta_2\mu)^2}{2(\beta_2^2\sigma_0^2+\sigma_2^2)}\right),\end{aligned}$$

and the second part is

$$\begin{aligned}&-\log P_{\text{GP}}(B|A) + \log P_{\text{GP}}(A|B)\\ =&\tfrac{n}{2}\log\left(\frac{\beta_2^2\sigma_0^2 + 2\frac{\beta_2^2}{\alpha_2^2}\sigma_1^2 + \sigma_2^2}{\alpha_2^2\sigma_0^2 + 2\frac{\alpha_2^2}{\beta_2^2}\sigma_2^2 + \sigma_1^2}\right) + \sum_{i=1}^{n}\left\{\frac{(b_i - (\beta_1 - \frac{\alpha_1}{\alpha_2}\beta_2) - \frac{\beta_2}{\alpha_2}(\alpha_1+\alpha_2\mu))^2}{2(\beta_2^2\sigma_0^2 + 2\frac{\beta_2^2}{\alpha_2^2}\sigma_1^2 + \sigma_2^2)}\right.\\ &\left. - \frac{(a_i - (\alpha_1 - \frac{\beta_1}{\beta_2}\alpha_2) - \frac{\alpha_2}{\beta_2}(\beta_1+\beta_2\mu))^2}{2(\alpha_2^2\sigma_0^2 + 2\frac{\alpha_2^2}{\beta_2^2}\sigma_2^2 + \sigma_1^2)}\right\}.\end{aligned}$$

The gap Δ decreases when σ_0 is relatively large, and when α_2 is close to β_2. Moreover, Δ increases linearly with the data size n.